



GraphShield 730

Micronized composite of synthetic wax and edge functionalized graphene (EFG) for improved anticorrosion in powder coatings

Features and Benefits

- Improves corrosion resistance on bare and treated cold rolled steel (CRS) when formulated with titanium dioxide or other inorganic pigments
- Black wax composite delivers graphene in an easy to use micronized powder
- Formulated with Asbury Carbons EFG technology
- 1% typically reduces mm creep by 25%
- Must be added to the premix, ideally processed above 110 ° C

Composition

Graphene/synthetic wax

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Powder coatings, industrial coatings.

Typical Properties*

	<u>GraphShield 730</u>
Mean Particle Size (µm)	8.0 - 12.0
Maximum Particle Size (µm)	31.00
Melting Point °C	108 - 113
Density @ 25 °C (g/cc)	1.24

Mar-26



MP-22

High performance, versatile grade of micronized synthetic wax for improved slip, scratch and rub resistance in a wide variety of paints, coatings and inks

Features and Benefits

- Imparts a high degree of lubricity and slip with rub and mar resistance
- Provides antiblocking properties
- Gives long-lasting water beading and weather resistance to exterior alkyd-based stains
- Consider MP-22C & MP-28C (spherical) for improved surface slip
- Biodegradable to OECD 302C test standard
- Not a microplastic per ECHA definitions

Composition

Synthetic wax

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and masonry); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; interior and exterior can and container coatings; coil coatings; powdered metals.

Typical Properties*

	<u>MP-22</u>	<u>MP-22VF</u>	<u>MP-22XF</u>	<u>MP-22XXF</u>
Mean Particle Size (µm)	7.0 - 10.0	6.0 - 8.0	4.5 - 6.5	3.75 - 5.75
Maximum Particle Size (µm)	31.00	22.00	22.00	15.56
Melting Point °C	110 - 115	110 - 115	110 - 115	110 - 115
Density @ 25 °C (g/cc)	0.93	0.93	0.93	0.93
NPIRI Grind	4.0 - 6.0	2.0 - 3.5	1.5 - 3.0	1.0 - 2.3

This product is also available as a water based wax dispersion - Microspersion 22-50

Mar-26

PolyGlide 1226XF

A highly engineered HDPE/ceramic/nanoceramic composite for abrasion resistance and lubricity

Features and Benefits

- HDPE composite reinforced with hard, inert ceramic microspheres and nanoceramic platelets
- Improved Taber abrasion resistance when compared to PE/PTFE additives
- Provides slip and lubricity
- Ideal for can and container coatings; 21CFR 175.300 approved
- Effective replacement for PTFE additives
- Compare to (coarser) Polyfluor 900

Composition

HDPE/ceramic

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

Typical Properties*

	<u>PolyGlide 1226XF</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	109 - 115
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 1226XF-50

Mar-26

MicroKlear 418

Micronized T-1 prime yellow refined carnauba wax for surface slip and scratch resistance with excellent clarity and gloss retention. Also useful as an additive in seed coatings

Features and Benefits

- Provides surface lubricity and scratch resistance
- Does not adversely affect clarity or gloss
- 100% natural and biodegradable
- Easy to disperse fine powder that can be incorporated with high speed mixing
- In seed coatings, improves anti-blocking, flow, and abrasion resistance

Composition

Carnauba wax

Renewable Carbon Index

100%

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and leather); wood coatings; printing inks and OPV's (including flexo and gravure); interior and exterior can and container coatings; coil coatings; seed coatings.

Typical Properties*

	<u>MicroKlear 418</u>
Mean Particle Size (µm)	6.0 - 8.0
Maximum Particle Size (µm)	22.00
Melting Point °C	81 - 86
Density @ 25 °C (g/cc)	1.00
NPIRI Grind	2.0 - 3.5
Color	Light Yellow

This product is also available as a water based wax dispersion - Microspersion 418-40

Mar-26

AquaBead® 519

A specially formulated wax composite that provides immediate water repellency and weather resistance

Features and Benefits

- Imparts immediate water beading
- Provides long-lasting water repellency
- Contains lower molecular weight polymers for faster water beading in water based and solvent based penetrating stains
- Combine with silica to improve weather resistance

Composition

Hydrophobically modified synthetic wax

Recommended Addition Levels

1.0-4.0% (on total formula weight)

Systems and Applications

Water based and solvent based coatings. Industrial coatings (including metal and masonry); stains, sealers and varnishes; wood coatings; automotive polishes.

Typical Properties*

	<u>AquaBead 519</u>
Mean Particle Size (µm)	5.5 - 8.0
Maximum Particle Size (µm)	22.00
Melting Point °C	126 - 132
Density @ 25 °C (g/cc)	0.94
NPIRI Grind	2.0 - 3.0

This product is also available as a water based wax dispersion - Microspersion 519

Mar-26

Superslip 6515AL

A highly engineered HDPE/EBS/aluminum oxide nanocomposite for maximum scratch, scuff and block resistance

Features and Benefits

- Maximum scratch resistance; equal to PE/PTFE additives
- HDPE/EBS composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Good abrasion resistance with slip and lubricity
- Amide wax component provides excellent antiblocking and release properties
- Ideal for can and container coatings; 21CFR 175.300 approved

Composition

HDPE/EBS/aluminum oxide

Renewable Carbon Index

>90%

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; can, container, and coil coatings; rubber additives.

Typical Properties*

	<u>Superslip 6515AL</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	139 - 145
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 6515AL-40

Mar-26



MPP-620

High performance micronized polyethylene wax which provides a unique combination of surface slip, abrasion/rub resistance, and antiblocking in a wide variety of coatings and inks

Features and Benefits

- Imparts excellent abrasion, rub and mar resistance with good surface slip
- Provides excellent antiblocking properties
- Adds heat resistance
- Better resistance to solvent absorption and swelling when compared to Synthetic wax

Composition

High density polyethylene

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; interior and exterior can and container coatings; coil coatings.

Typical Properties*

	<u>MPP-620VF</u>	<u>MPP-620XF</u>	<u>MPP-620XXF</u>
Mean Particle Size (µm)	5.0 - 7.0	4.5 - 5.5	4.25 - 4.75
Maximum Particle Size (µm)	22.00	22.00	12.00
Melting Point °C	120 - 124	120 - 124	120 - 124
Density @ 25 °C (g/cc)	0.96	0.96	0.96
NPIRI Grind	2.0 - 3.0	1.0 - 2.0	1.0 - 1.5

This product is also available as a water based wax dispersion - Microspersion 620VF-50

Mar-26

MicroKlear 418AL

Aluminum oxide modified carnauba wax for maximum scratch resistance with surface slip, clarity and gloss retention. Also useful as an additive in seed coatings

Features and Benefits

- Carnauba wax composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Provides maximum scratch resistance with surface lubricity
- Does not adversely affect clarity or gloss
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite wax/aluminum oxide particle is less abrasive on processing equipment compared to free aluminum oxide
- Effective replacement for PTFE additives

Composition

Carnauba wax/aluminum oxide

Renewable Carbon Index

>95%

Recommended Addition Levels

1.0-3.0% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic, metal and leather); wood coatings; printing inks and OPV's (including flexo and gravure); interior and exterior can and container coatings; coil coatings; seed coatings.

Typical Properties*

	<u>MicroKlear 418AL</u>
Mean Particle Size (µm)	6.0 - 8.0
Maximum Particle Size (µm)	22.00
Melting Point °C	81 - 86
Density @ 25 °C (g/cc)	1.04
NPIRI Grind	2.0 - 3.5

This product is also available as a water based wax dispersion - Microspersion 418AL-40

Mar-26

MPP-620AL-XF

A highly engineered polyethylene/aluminum oxide nanocomposite that provides slip, abrasion/rub resistance, and antiblocking

Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

Composition

High density polyethylene/aluminum oxide

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

Typical Properties*

	<u>MPP-620AL-XF</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	120 - 124
Density @ 25 °C (g/cc)	1.00
NPIRI Grind	1.0 - 2.0

Mar-26

MPP-611AL

A highly engineered HDPE/aluminum oxide nanocomposite for unsurpassed scratch and scuff resistance

Features and Benefits

- Maximum scratch and scuff resistance with slip and lubricity
- HDPE composite reinforced with 300 nm aluminum oxide nanoparticles (Mohs Hardness 9)
- Easier-to-disperse nanotechnology in a safe, non-nano powder
- Composite particle density maximizes efficiency and minimizes settling
- Effective replacement for PTFE additives

Composition

HDPE/aluminum oxide

Recommended Addition Levels

0.5-1.5% (on total formula weight)

Systems and Applications

Water based, solvent based and energy curable coatings and inks. Industrial coatings (including plastic and metal); stains, sealers and varnishes; wood coatings; printing inks and OPV's (including flexo and gravure); powder coatings; coil coatings; rubber additives.

Typical Properties*

	<u>MPP-611AL</u>
Mean Particle Size (µm)	3.5 - 5.5
Maximum Particle Size (µm)	15.56
Melting Point °C	110 - 116
Density @ 25 °C (g/cc)	0.99
NPIRI Grind	1.0 - 2.0

This product is also available as a water based wax dispersion - Microspersion 611AL-50

Mar-26